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Extensive transcriptional variation poses a challenge to thermal stress biomarker development for endangered corals

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Abstract

As climate changes, sea surface temperature anomalies that negatively impact coral reef organisms continue to increase in frequency and intensity. Yet, despite widespread coral mortality, genetic diversity remains high even in those coral species listed as threatened. While this is good news in many ways, it presents a challenge for the development of biomarkers that can identify resilient or vulnerable genotypes. Taking advantage of three coral restoration nurseries in Florida that serve as long-term common garden experiments, we exposed over 30 genetically distinct *Acropora cervicornis* colonies to hot and cold temperature shocks seasonally and measured pooled gene expression responses using RNAseq. Targeting a subset of 20 genes, we designed a high-throughput qPCR array to quantify expression in all individuals separately under each treatment with the goal of identifying predictive and/or diagnostic thermal stress biomarkers. We observed extensive transcriptional variation in the population, suggesting abundant raw material is available for adaptation via natural selection. However, this high variation made it difficult to correlate gene expression changes with colony performance metrics such as growth, mortality and bleaching susceptibility. Nevertheless, we identified several promising diagnostic biomarkers for acute thermal stress that may improve coral restoration and climate change mitigation efforts in the future.

KEYWORDS

conservation genetics, coral reef, gene expression, restoration, *Symbiodinium*, symbiosis

1 | INTRODUCTION

Colonies of the branching staghorn coral, *Acropora cervicornis*, once formed dense thickets along shallow reef zones throughout the entire Caribbean region. Due to anthropogenic impacts, disease and temperature-induced bleaching events, the abundance of this prevalent species has declined by more than 80% in recent decades (Bruckner, 2002; Dudgeon, Aronson, Bruno, & Precht, 2010), and it is currently listed as threatened under the U.S. Endangered Species Act. In an effort to conserve and restore declining populations, coordinated coral propagation and reef restoration efforts have been

developed throughout the Caribbean (reviewed by Young, Schopmeyer, & Lirman, 2012). Most programmes propagate corals within in-water nurseries that serve as common gardens, where genetically diverse colonies are reared in close proximity for extended periods before being planted back onto degraded reefs. These nurseries function as active restoration tools as well as genetic repositories that protect diversity during stress events (Schopmeyer et al., 2012).

A guiding principle of coral conservation is to enhance the genetic diversity of dwindling populations and thus improve resilience by preserving varied stress responses among individuals (Baums, 2008). This principle was followed when designing the

Caribbean restoration programmes by selecting donor colonies that were distinct at neutral microsatellite markers (Baums, Hughes, & Hellberg, 2005), translating into high genomic variation within and between sites (Drury et al., 2016; Drury, Schopmeyer, et al., 2017). Additionally, functional variation in the nurseries is high (Lirman et al., 2014; Lohr & Patterson, 2017), meaning many *A. cervicornis* genotypes perform differently in the same environment or under the same stress (Lirman et al., 2011).

Climate change is expected to increase the magnitude and frequency of hot and cold sea surface temperature anomalies in their respective seasons (Easterling et al., 2000), especially in high-latitude marginal regions such as the Florida Keys, USA (Boesch, Field, & Scavia, 2000). Both hot and cold events are known to drive coral bleaching (symbiont loss) and mortality (Jokiel & Coles, 1990; LaJeunesse, Reyes-Bonilla, & Warner, 2007; Saxby, Dennison, & Hoegh-Guldberg, 2003). However, hot and cold events impact different physiological mechanisms (Roth, Goericke, & Deheyn, 2013), and tolerance trade-offs may exist. For example, coral mortality was rampant during the extreme 2010 Florida Keys cold water event, which disproportionately affected inshore colonies previously resilient to summer hot water bleaching (Kemp et al., 2011, 2016; Lirman et al., 2011). Such patterns suggest that hot and cold stresses require unique physiological and molecular responses, implying that corals may have to gain multiple adaptations to expand their ranges poleward where climate variability is larger.

The health of a coral colony is also tied to the identities and physiological qualities of its dinoflagellate endosymbionts (Sampayo, Ridgway, Bongaerts, & Hoegh-Guldberg, 2008); together the host and all of its microbial symbionts constitute the coral holobiont. Mature *A. cervicornis* holobionts typically associate with just one *Symbiodinium* species (*S. "fitti"* = ITS2 type A3^{Caribbean}) at depths above 10 m (Thornhill, LaJeunesse, Kemp, Fitt, & Schmidt, 2006). Moreover, most of the population of symbiont cells within a colony comprises one clonal cell line (strain), similar to symbiont populations observed in colonies of a related host species, *A. palmata* (Baums, Devlin-Durante, & LaJeunesse, 2014; Thornhill, Howells, Wham, Steury, & Santos, 2017). However, low-abundance background symbionts from other *Symbiodinium* "clades" have been detected in many *A. cervicornis* colonies (Baums, Johnson, Devlin-Durante, & Miller, 2010; Silverstein, Correa, & Baker, 2012). *Symbiodinium* clade identity can affect host transcription in corals (DeSalvo et al., 2010), but it is unclear whether within-species diversity in the dominant symbiont and/or variation in the abundance of background

symbionts have similar effects, although theory predicts they could (Parkinson & Baums, 2014; Ziegler, Eguíluz, Duarte, & Voolstra, 2017).

Genetic variation that correlates with environmental conditions or thermal tolerance can be developed as coral biomarkers (Jin et al., 2016; Lundgren, Vera, Peplow, Manel, & van Oppen, 2013). Gene expression biomarkers (GEBs), which capture dynamic stress responses, are promising tools for coral restoration (reviewed by Louis, Bhagooli, Kenkel, Baker, & Dyall, 2016). GEBs fall into two major categories: those that predict future performance and those that diagnose current health status. By characterizing the molecular profiles that correspond to stress tolerance and stress sensitivity, it is possible to monitor the health of reef organisms, assess acute anthropogenic impacts (e.g., pollution effects) and identify resilient genotypes for propagation in coral nurseries. However, the genetic architectures of coral traits are largely unknown, adding to the challenge of developing biomarkers. The initial analyses of GEBs have been restricted in the number of host genes investigated, the number of individuals assessed, and the time frame over which colonies have been sampled, all without detailed knowledge of the resident symbionts. Here, we use high-throughput molecular approaches to expand the scope of *A. cervicornis* GEB development and quantify gene expression variation in a coral population from a marginal high-latitude habitat.

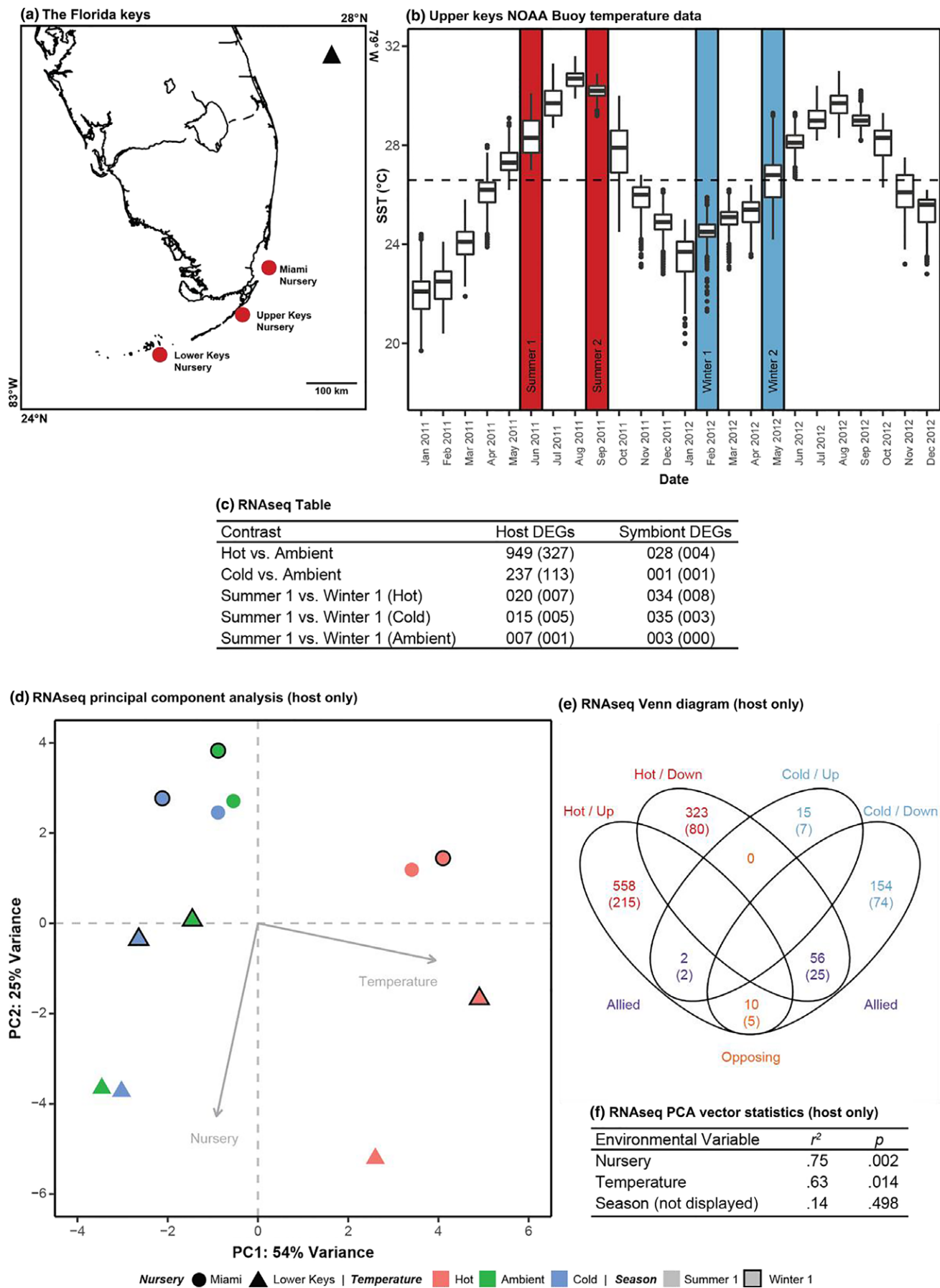
2 | MATERIALS AND METHODS

For full methodological details, see Appendix S1. All raw data, R code and additional supplements can be accessed in the Pennsylvania State University's SCHOLARSphere database (<https://doi.org/10.18113/s1rp4r>). The RNAseq data are also deposited in the NCBI GEO database (Accession no. GSE108338).

2.1 | Coral nurseries and colony selection

Three in-water coral propagation nurseries in Florida were targeted for this experiment: the University of Miami nursery in Miami (referred to as the Miami nursery; 25°28'24.24"N, 80°07'42.24"W), the Coral Restoration Foundation nursery in Tavernier (Upper Keys nursery; 24°58'55.84"N, 80°26'10.69"W) and the MOTE Marine Lab/Nature Conservancy nursery in Summerland Key (Lower Keys nursery; 24°33'45.68"N, 81°24'00.54"W) (Figure 1a). These three locations fall within the biogeographic range of one intermixing *Acropora*

FIGURE 1 RNAseq experimental overview. (a) Map of the Florida Keys depicting the location of the three nurseries used in the study. (b) Monthly sea surface temperature observations (SST) near the Upper Keys nursery from 2011–2012 (data from NOAA National Data Buoy Center; <http://www.ndbc.noaa.gov/>). Bars indicate sampling points, with red corresponding to Summer and blue corresponding to Winter. (c) The number of differentially expressed genes (DEGs) among coral hosts and algal symbionts for each main contrast in the RNAseq experiment (left value: total number of DEGs; right value in parentheses: number of well-annotated DEGs). (d) Principal component analysis (PCA) of all host DEGs with significant environmental vectors superimposed. The vectors point in the direction of the greatest change in the variable of interest, while the vector length is proportional to the correlation between the ordination and the variable. (e) Venn diagram depicting overlap in the total number of host DEGs (top value) or well-annotated DEGs (bottom value in parentheses) that were up- or down-regulated in the hot or cold treatment relative to ambient conditions. Overlapping regions correspond to allied or opposing expression patterns. (f) Statistics for the PCA environmental vectors [Colour figure can be viewed at wileyonlinelibrary.com]



cervicornis population near the northernmost boundary of the species (Baums et al., 2010; Drury et al., 2016). All colonies had been growing in their respective common garden nurseries for at least 1 year prior to experimentation, minimizing environmental variation. At the Miami nursery, the annual linear tissue extension rates, annual mortality rates among replicate colonies and relative bleaching sensitivities for each genotype were monitored throughout the experimental period.

As part of the initial collection when the nurseries were established, donor colonies were genotyped at four host-specific microsatellite loci (Baums et al., 2005) to establish each colony's genet identity. Subsequently, the dominant *Symbiodinium* "fitti" strain in each colony was genotyped at 13 symbiont-specific loci (Baums et al., 2014; Pinzon, Devlin-Durante, Weber, Baums, & LaJeunesse, 2011). Host genets with anecdotally variable growth rates, mortality rates and bleaching susceptibilities were chosen from each nursery (Miami: $n = 10$; Upper Keys: $n = 11$; Lower Keys: $n = 10$; in total, 31 genets). Each collection included seven unique *S. "fitti"* strains. No host genets or symbiont strains were shared across nurseries (Table S1). Host genotypes were not verified during the experiment because so many samples were collected, but the probability of misidentification was low due to the way genets are organized and propagated within the nurseries.

To examine intra-individual variation, three of the 31 host genets were subsampled an additional two times each from physically separate colonies (three ramets of the same genet), so a total of 37 colonies were included in the study. To examine the effect of genotypic variation in the dominant symbiont, collections at each nursery included at least three genetically distinct colonies each associating with the same *S. "fitti"* strain ("monotypic" group), as well as three colonies each associating with unique *S. "fitti"* strains ("diverse" group).

2.2 | Temperature experiments

A single branch (~9 cm) from each colony at each nursery was clipped underwater via SCUBA. Corals were sampled within 2 hr of solar noon, and the growing tips were removed to minimize effects of diel cycle and branch position on gene expression (Hemond & Vollmer, 2015). The remaining branch was divided into three equal fragments, which were exposed for 1 hr to one of three treatments: extreme hot (35°C), extreme cold (10°C), or ambient (season-dependent: 24–28°C). Treatments were conducted shipboard immediately after collection. Temperatures were maintained in insulated water buckets using temperature regulators, aquarium heaters and/or ice as needed. The buckets were covered to block light. This set-up was designed to be relatively inexpensive, rapid and simple for restoration workers to repeat in the future (a schematic is provided in Figure S1).

To keep track of genotypes, fragments were maintained separately in labelled plastic bags sealed with water from the treatment bucket. All bags were submerged simultaneously in the buckets and stirred periodically to equalize temperature. There was no flow or

oxygen exchange within each bag. In the ambient treatment, *A. cervicornis* did not show signs of stress under such conditions for the short exposure time. Based on an instant-read thermometer, the design maintained target temperatures for the 1 hr treatment duration. After the exposure, fragments were preserved in RNALater.

The experiment was repeated four times over a 12-month period: June 2011 (Summer 1), September 2011 (Summer 2), February 2012 (Winter 1) and May 2012 (Winter 2; Figure 1b). The Summer collections were scheduled to capture gene expression before and after the Summer thermal maximum. Due to logistical issues, it was impossible to sample around the Winter thermal minimum in the same way. Instead, the Winter collections began immediately after the Winter thermal minimum and spanned the same time interval as the Summer collections.

Our treatment temperatures were extreme, as bleaching and mortality are typically observed at <16°C and >31°C in this species, and it is unlikely that any coral would experience such dramatic instantaneous temperature changes naturally. An alternative approach would have been to bring the coral fragments to a laboratory to acclimatize them using more natural temperatures. However, the re-establishment of an aquarium-based common garden would negate the power provided by the field-based nursery (namely, the myriad factors—both known and unknown—that influence survival in the wild and cannot be replicated in an aquarium). The immediate short and extreme stress treatment was thus the best approach to accentuate transcriptomic response differences among genotypes.

2.3 | RNAseq experiment

A total of 444 *A. cervicornis* experimental fragments were collected from the Florida coral nurseries (37 colonies $\times$ 3 temperature treatments $\times$ 4 seasons). First, an RNAseq experiment was carried out on pooled samples to identify important stress response genes that could later be assayed in all samples using high-throughput qPCR (HTqPCR). The subset included seven host genets each from two of the three nurseries (Miami and Lower Keys), two of the four seasonal collections (Summer 1 and Winter 1) and all three temperature treatments (ambient, hot and cold), resulting in 12 libraries.

Total RNA extraction, library preparation, Illumina sequencing, read processing, functional annotation and differential expression analyses were carried out as by Parkinson et al. (2016) with minor modifications (Appendix S1), resulting in ~37 million high-quality reads per library. Reads were mapped to the *A. cervicornis* transcriptome of Libro, Kaluziak, and Vollmer (2013) and separated into host and symbiont components bioinformatically using additional coral and *Symbiodinium* genomic resources (Appendix S1). The host gene set ($n = 22,772$) was analysed separately from the symbiont gene set ($n = 21,094$). Transcripts with unassigned or ambiguous origin ($n = 21,669$) were excluded from further analysis. All genes were modelled in the R package EDGER (Robinson, McCarthy, & Smyth, 2010) and assessed for differential expression with respect to temperature treatment (season ignored) and season (temperatures analysed separately) using nurseries as

replicates. Results were visualized through principal component analysis and Venn diagrams. Additional visualizations (biological coefficient of variation, multidimensional scaling, mean expression vs. variance and volcano plots) are available for each contrast in the SCHOLARSPHERE database.

2.4 | HTqPCR array design

The RNAseq experiment yielded the targets for the HTqPCR array. The array was developed to assess expression separately for each individual coral, in contrast to the pooled design of the RNAseq experiment. The TaqMan OpenArray platform (Thermo Fisher Scientific, Waltham, MA, USA) accommodated 28 targets in duplicate per plate. Based on the RNAseq experiment, it appeared that certain host genes responded in the same direction under both hot and cold stresses (allied pattern), others responded in opposite directions (opposing pattern), still others responded to only one stress (uncoupled pattern) and some varied across time (seasonal pattern).

We chose 21 functionally relevant host genes from each expression category along with three low-variance host control genes and four clade-specific *Symbiodinium* ribosomal genes for the HTqPCR array. The symbiont genes were used to track abundances of Clades A, B, C and D, but no further *Symbiodinium* genes were included due to the low incidence of differential expression. Host genes were chosen based on meeting a majority of several selection criteria (Appendix S1). Ideally they were annotated, differentially expressed, functionally enriched, intron spanning, universal (similar expression across nurseries), and part of a temperature coexpression module. Complete gene IDs, array order, annotation information, response categories and RNAseq expression patterns for each target are presented in Figure 2a.

2.5 | HTqPCR array experiment

After choosing genes of interest, all 28 targets were printed in duplicate (for technical replication) on 10 TaqMan OpenArray custom plates, accommodating the 444 samples (24,864 unique qPCR reactions). Total RNA from each sample (100 ng) was treated with DNase I to remove gDNA, converted to cDNA with a High-Capacity cDNA Reverse Transcription kit and pre-amplified using TaqMan Custom PreAmp Pools (all kits from Thermo Fisher Scientific). qPCRs were performed on a QuantStudio 12K Flex Real-Time PCR System at the Pennsylvania State University Genomics Core Facility.

Raw cycle threshold (C_T) values for each qPCR were processed in R with the package MCMC.QPCR (Matz, Wright, & Scott, 2013). After accounting for differences in amplification efficiencies and removing outliers, these values were fit to the "classic" linear mixed model, which uses Markov Chain Monte Carlo simulations for maximum-likelihood analysis. The model specifically tested the fixed effects of temperature, season and their interaction on each gene, normalized to the control genes and incorporating a random effect of host genotype. One gene target (A17_Gnat3) had very low efficiency, multiple cases of nonamplification and highly variable expression

values; therefore, it was dropped from the model and further consideration. A separate time-series analysis was performed using the R package MASIGPRO (Conesa, Nueda, Ferrer, & Talón, 2006) on the averaged, normalized relative expression values generated with MCMC.QPCR.

2.6 | Visualizing host gene expression variation

To visualize gene expression variation among host genets, DataAssist (Thermo Fisher Scientific) was used to remove outlier C_T values, calculate mean C_T values based on the technical replicates for each sample and normalize to the average expression of the three endogenous control genes (dC_T). The values were then imported into the R statistical environment, where expression relative to ambient temperature control samples (ddC_T) was calculated for each gene separately and plotted on a $\log_2$ scale.

To examine the effect of dominant *S. "fitti"* genotypic variation on host expression variation, "monotypic" versus "diverse" groups were compared. For each gene at each nursery and season, the variance in stress response values (ddC_T for hot or cold) across the three "monotypic" or three "diverse" colonies was calculated. Gene-wise differences in variance distributions between groups were assessed via nonparametric Mann–Whitney–Wilcoxon test ($\alpha = 0.05$).

2.7 | Background symbiont analysis

Typically, *Symbiodinium* from Clades B, C and D represent <5% of the symbiont community in Florida's shallow-water acroporid corals (Parkinson, Banaszak, Altman, LaJeunesse, & Baums, 2015; Thornhill et al., 2006). Special consideration was given to the quantification of background symbiont types, which was based on the count of symbiont-specific reads. The PreAmp Pools included primers for all targets except the Clade A *Symbiodinium* ribosomal gene, as this target already yielded very high signal because of the overwhelming numerical dominance of *S. "fitti"* in all samples. Pre-amplification would have potentially negatively influenced the efficiency of other reactions. Thus, the Clade A values were not directly comparable to those of Clades B, C and D and were excluded from the analyses below.

2.8 | Correlation analysis

To explore correlations among expression profiles (both raw expression levels and fold-change responses relative to ambient), background symbiont abundances (*Symbiodinium* Clades B, C and D) and physiological metrics (growth rate, mortality and bleaching), all relevant data from the Miami nursery were combined in a single matrix. Growth was measured as annual linear extension averaged over all ramets of a genet from 2011 to 2012, mortality was measured as annual per cent mortality among ramets of a genet from 2011 to 2012, and bleaching was a qualitative measure of susceptibility during a later bleaching event in 2014 (categorized as 1 = no bleaching,

(a) HTqPCR assay table

Assay ID	Gene ID	Origin	Category	Expression pattern	Genbank	Uniprot	Description
A01	Shell	Host	Allied	Hot Up Cold Up	GASU01084441	P86982	Insoluble matrix shell protein
A02	GXN	Host	Allied	Hot Up Cold Up	GASU01031661	D9IQ16	Galaxin
A03	CLEC4	Host	Allied	Hot Down Cold Down	GASU01085638	Q9ULY5	C-type lectin domain family 4 member E
A04	ALKBH1	Host	Allied	Hot Down Cold Down	GASU01070780	T2ME17	Alkylated DNA repair protein alkB homolog
A05	Drip	Host	Allied	Hot Down Cold Down	GASU01030213	Q9V5Z7	Aquaporin
A06	GLIPR2	Host	Allied	Hot Down Cold Down	GASU01080687	Q9H4G4	Golgi-associated plant pathogenesis-related protein
A07	TRAF3	Host	Allied	Hot Down Cold Down	GASU01030289	Q13114	TNF receptor-associated factor 3
A08	sno1	Host	Opposing	Hot Up Cold Down	GASU01086826	Q8MP06	Senecionine N-oxygenase
A09	Greb1l	Host	Opposing	Hot Up Cold Down	GASU01071194	B9EJV3	GREB1-like protein
A10	fgfr1a	Host	Opposing	Hot Up Cold Down	GASU01030182	Q90Z00	Fibroblast growth factor receptor 1-A
A11	hsp16.2	Host	Opposing	Hot Up Cold Down	GASU01030017	P06582	Heat-shock protein Hsp-16.2
A12	ZFAND2B	Host	Opposing	Hot Up Cold Down	GASU01081142	Q8WV99	AN1-type zinc finger protein 2B
A13	RTKN	Host	Uncoupled	Hot Up Only	GASU01083711	Q9BST9	Rhotekin
A14	GADD45A	Host	Uncoupled	Hot Up Only	GASU01080585	Q3ZBN6	Growth arrest and DNA damage-inducible protein
A15	wnt4	Host	Uncoupled	Hot Down Only	GASU01049477	P49338	Protein Wnt-4
A16	ACP5	Host	Uncoupled	Hot Down Only	GASU01049261	P09889	Tartrate-resistant acid phosphatase type 5
A17	Gnat3	Host	Uncoupled	Cold Up Only	GASU01040349	P29348	Guanine nucleotide-binding protein
A18	sesn1	Host	Uncoupled	Cold Down Only	GASU01083762	P58003	Sestrin-1
A19	TIR1	Host	Uncoupled	Cold Down Only	GASU01040585	Q15399	Toll-like receptor 1
A20	EXD1	Host	Seasonal	Winter Down	GASU01070610	Q8NHP7	Exonuclease 3'-5' domain-containing protein
A21	ANKRD44	Host	Seasonal	Winter Down	GASU01040662	Q5F478	Serine/threonine-protein phosphatase 6
A22	Mob3	Host	Host Control	Constant	GASU01031038	Q9VL13	MOB kinase activator-like protein
A23	SEPT7	Host	Host Control	Constant	GASU01084761	Q08DM7	Neuronal-specific septin
A24	ACT7	Host	Host Control	Constant	GASU01085644	P53492	Actin-7
A25	SymA_28S	Symbiont	Clade A Specific	NA	KF364601	NA	Large Subunit rRNA
A26	SymB_28S	Symbiont	Clade B Specific	NA	KT149345	NA	Large Subunit rRNA
A27	SymC_28S	Symbiont	Clade C Specific	NA	FJ529523	NA	Large Subunit rRNA
A28	SymD_28S	Symbiont	Clade D Specific	NA	KF740689	NA	Large Subunit rRNA

(b) HTqPCR average expression response

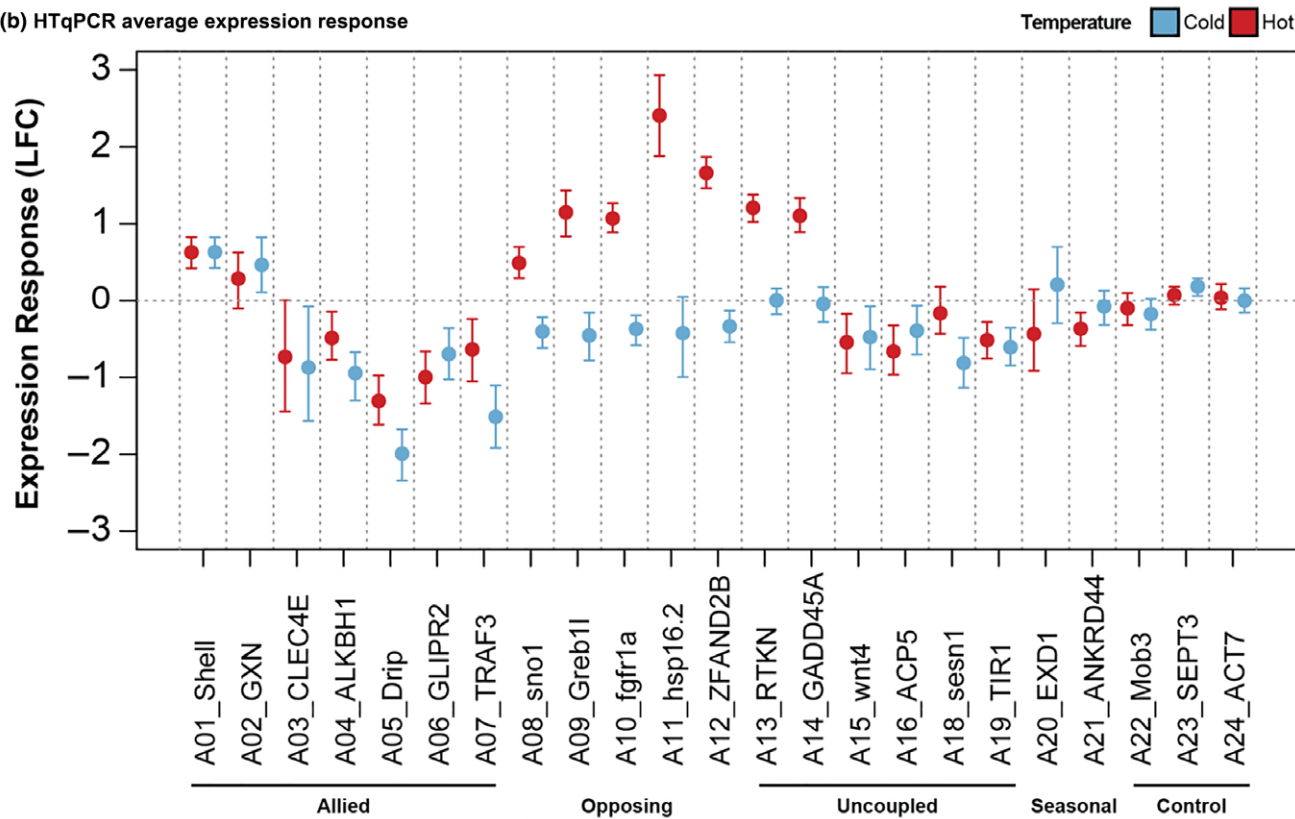


FIGURE 2 HTqPCR experimental overview. (a) Table of gene targets for the high-throughput qPCR (HTqPCR) experiment, including annotation information and expected expression patterns based on the RNAseq experiment. (b) Thermal shock responses for coral host genes included on the HTqPCR array averaged across all genotypes and seasons. Points represent host expression responses ($\log_2$ fold changes; LFC) relative to ambient controls for cold (blue; $n = 146$ measurements per gene) and hot (red; $n = 145$ measurements per gene) temperature treatments. Error bars represent 95% credible intervals from the Bayesian linear mixed model. Significant divergence from 0 is evident when the bars do not cross the horizontal reference at $y = 0$; significant divergence between cold and hot treatments for a given gene is evident when the bars from different treatments do not cross each other [Colour figure can be viewed at wileyonlinelibrary.com]

2 = bleached during summer and recovered and 3 = bleached and died). Pearson correlation coefficients with a Holm multiple comparison correction were calculated pairwise ($\alpha = 0.05$). Calculations were performed in R, and correlation heatmaps were generated in Gene-E (Gould, 2015).

3 | RESULTS

3.1 | Global expression patterns (RNAseq)

The RNAseq experiment on a subset of samples and seasons showed temperature-based differential expression was more apparent in the coral host than the algal symbiont (e.g., 949 host differentially expressed genes (DEGs) vs. 28 symbiont DEGs in the heat shock vs. ambient treatment; Figure 1c). Few changes in gene expression for both hosts and symbionts were observed between Summer 1 and Winter 1. Given the low proportion of dynamic gene expression in *Symbiodinium*, we subsequently focused only on coral host genes. There were more host genes responsive to heat stress than cold stress (e.g., 949 hot DEGs vs. 237 cold DEGs). A Gene Ontology enrichment analysis revealed many DEGs were components of stress response processes (Table S2). In general, heat stress mostly resulted in gene upregulation, whereas cold stress mostly resulted in gene downregulation (Figure 1e).

Samples with similar host expression patterns were grouped visually through principal component analysis (Figure 1d,f). The first principal component accounted for 54% of expression variation and was largely correlated with temperature ($r^2 = .63$; $p = .014$). The second principal component accounted for 25% of variation and was correlated with nursery/location ($r^2 = .75$; $p = .002$). Season did not correlate with the ordination ($r^2 = .14$; $p = .498$). Many of the stress response genes appeared to show allied or opposing patterns (Figure 1e). Of the allied patterns, two genes were upregulated in both hot and cold, while 56 genes were both downregulated in hot and cold. Of the opposing patterns, 10 genes were upregulated in hot and downregulated in cold, while no genes were downregulated in hot and upregulated in cold.

3.2 | Targeted expression patterns (HTqPCR)

The HTqPCR experiment on a subset of 20 host genes showed the average expression of each gene under each treatment generally matched the pattern expected based on the RNAseq results. All allied genes responded to hot and cold stresses in the same direction, all opposing genes responded in opposite directions, and all control genes diverged very little from zero (Figure 2; Table S3a). According to the Bayesian model, most of these trends were statistically significant (or not significant, in the case of controls), and ultimately, 19 of the 23 genes analysed met significance expectations under both hot and cold conditions.

According to the separate time-series analysis (Figure 3 and Figure S2), nine of the 20 genes had expression profiles that varied with season under ambient conditions, while those nine plus an additional

three genes varied seasonally in terms of their responses to hot and cold (Table S3b), largely in agreement with the results of the Bayesian model. In the time-series model, none of the control genes were differentially expressed at any point or under any treatment. Interestingly, neither were the two genes that were expected to have a seasonal pattern based on RNAseq data (A20_EXD1 and A21_ANKRD44). Representative expression time series are presented in Figure 3, while plots for all genes can be found in Figure S2.

3.3 | Genotypic expression variation

Expression patterns were highly variable when each host genotype was considered separately. This is evident in the wide range of individual expression values for a given gene at each time point in Figure 3. An expanded example is provided in Figure 4 for target A12_hsp-16.2. While the hot treatment drove upregulation of A12_hsp-16.2 in all colonies, the extent of variation ranged from a log₂ fold change of ~1 to ~10 (or twofold to 1,024-fold). Notably, this range was observed among ramets of the same genet (colony U11 in Winter 1). If considered instead in terms of dominant symbiont genotype, results were similarly variable, with both high and low expressions among colonies sharing clonal S. "fitti" strains. Ultimately, no genes featured expression patterns that could be explained easily by host or dominant symbiont identity.

Nor was host expression related to dominant symbiont genotypic diversity. When comparing groups of colonies with identical host diversity but varying intraspecific symbiont diversity, no genes showed significant differences in host heat stress response variances. For the cold stress response, only three genes were significant (A05_Drip, A07_TRAF3 and A09_Greb1l), but in the opposite direction than expected (the "monotypic" variance was greater than the "diverse" variance). A typical example is given in Figure 6a for target A15_wnt4.

3.4 | Correlations of host expression with physiological data and background symbiont diversity

Host gene relative expression levels (dC_T) under ambient, hot and cold temperatures were generally unrelated to colony growth rates, mortality or bleaching categories (Figure 5), with typically low Pearson correlation coefficients ($|p| < .3$). Nevertheless, some correlations were quite strong and significant ($|p| > .7$; Figure 5a), but patterns were inconsistent across seasons. Results were similar for stress responses (ddC_T) and for nonparametric Spearman rank correlations (data not shown).

Background *Symbiodinium* abundance did not have an obvious effect on ambient host gene expression (Figure 6b), although four heat stress genes showed strong positive correlations with Clade D abundance during Summer 1 ($p > .7$). The relative abundance of different *Symbiodinium* did not predict host colony performance metrics at the Miami nursery consistently (Figure 6c), with the exception of a negative correlation between bleaching frequency and background symbiont abundance during Winter 2. While background Clade B, C

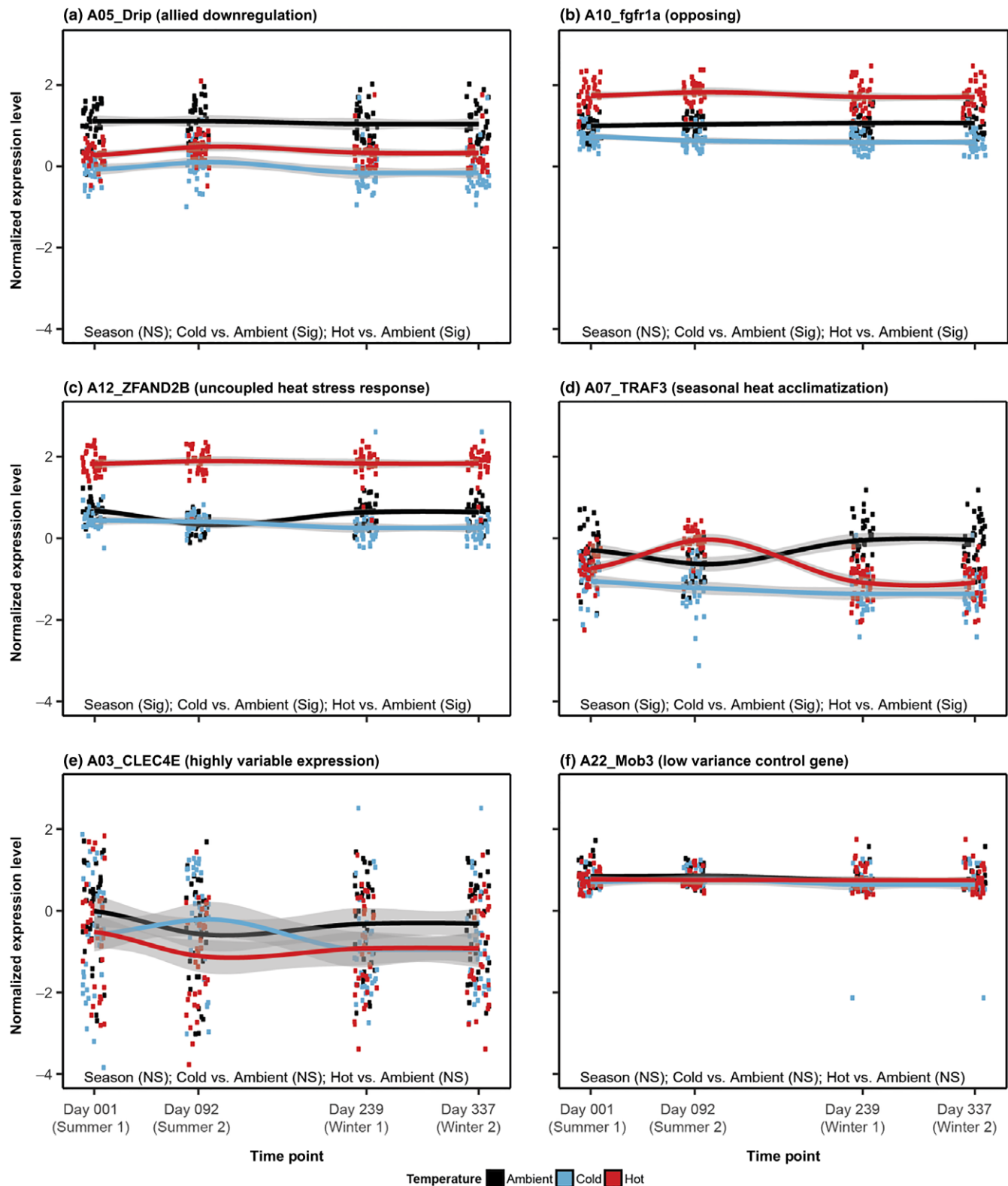


FIGURE 3 Host relative expression levels by season based on the HTqPCR experiment. Temperature treatment is indicated by color: ambient (black), cold (blue) and hot (red). Representative examples are provided for (a) allied downregulation, (b) opposing expression, (c) uncoupled heat stress response, (d) seasonal heat acclimatization, (e) highly variable expression and (f) low variance control genes. Expression patterns were explored with a time series analysis. Significance for the main effects of time (Season), cold stress (Cold vs. Ambient) and heat stress (Hot vs. Ambient) are indicated (NS = not significant; Sig = significant; $p < .05$). Gray shading corresponds to 95% confidence intervals [Colour figure can be viewed at wileyonlinelibrary.com]

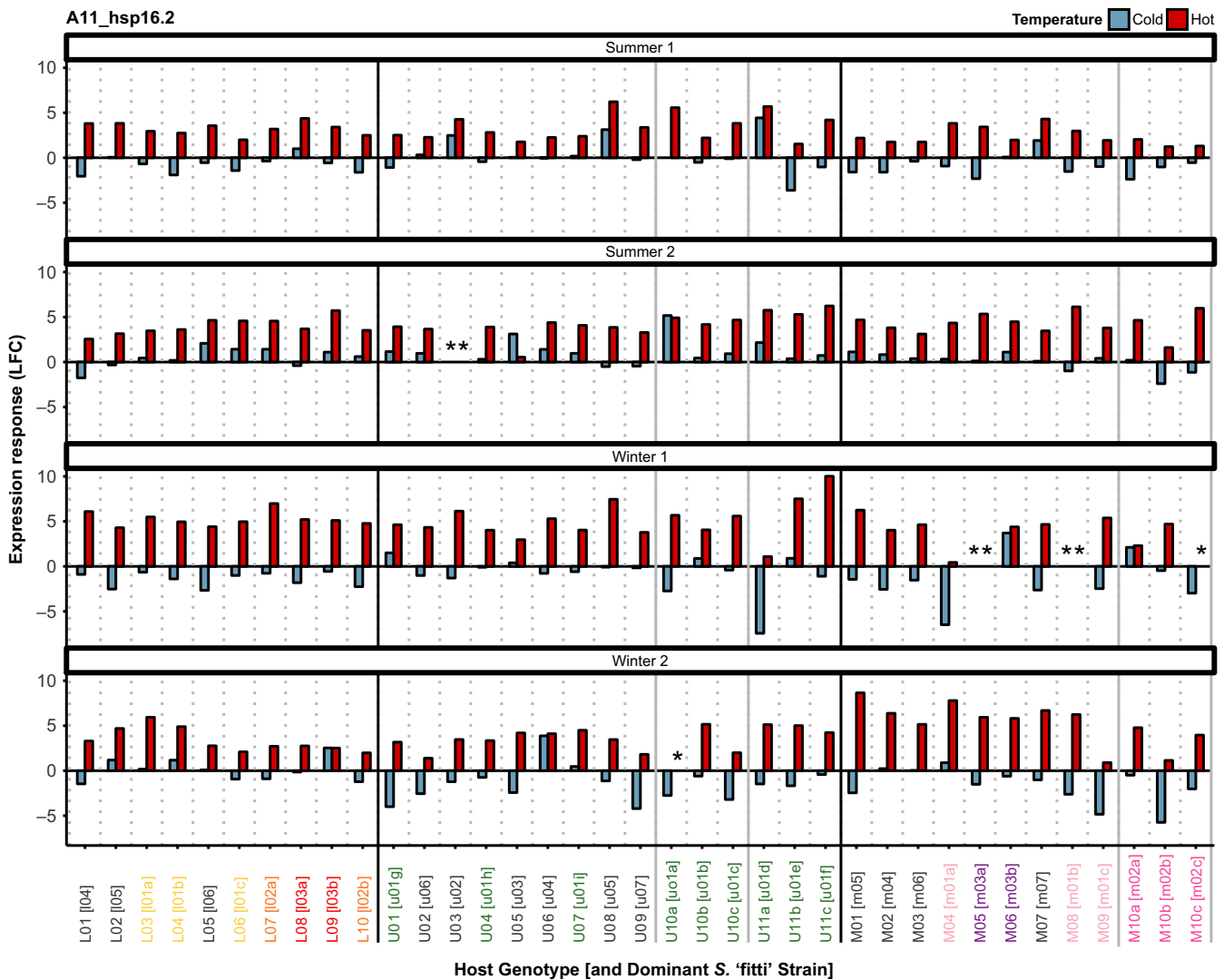


FIGURE 4 Relative expression variation (heat shock protein 16.2) among all colonies, nurseries and seasons. Bars represent host expression responses (log₂ fold changes; LFC) relative to ambient controls for cold (blue) and hot (red) temperature treatments. Colonies are grouped by nursery (separated by black vertical lines) and host genotypes along the x-axis. The first letter represents nursery (L = “Lower Keys”, U = “Upper Keys”, M = “Miami”), while numbers identify unique genotypes within each nursery. Dominant symbiont (*Symbiodinium* “fitti”) strain identities are also provided in brackets using lower case letters to indicate nursery and numbers for unique genotypes within a nursery (independent from host labels). Where expression values were determined for three replicate ramets of the same host genet, the values for each colony are plotted separately (but grouped together between solid gray lines) and identified by the same host genotype name appended with “a,” “b” and “c” suffixes. Clonal symbiont strains are also identified with shared names labeled with suffixes, as well as shared text color (unique strains are labeled in black). Colonies U03, U10a, M05, M08 and M10c are missing some data due to failed amplification, as indicated by asterisks [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

and D symbionts were detected in most colonies at most time points, the time-series analysis revealed no seasonal influence on their relative abundances.

4 | DISCUSSION

4.1 | Extensive expression variation among and within coral colonies

We observed a high degree of gene expression variation among *Acropora cervicornis* colonies, similar to other acroporid corals (e.g., Granados-Cifuentes, Bellantuono, Ridgway, Hoegh-Guldberg, &

Rodriguez-Lanetty, 2013; Parkinson et al., 2015). The average biological coefficient of variation—a measure of gene expression variation among biological replicates in an RNAseq study—was relatively high (~0.6 in most contrasts, compared to a typical value of 0.4 among independent biological samples; Chen, Lun, & Smyth, 2014). Host genotype, nursery of origin, stress type and time point all contributed to the diverse expression patterns (Figures 1–4). Remarkably, the expression of a given gene in physically separate colonies of the same clone could vary up to 1,024-fold despite shared proximity, environmental history and genetic make-up. Minor differences in handling, experimental treatment, circadian cycle, microhabitat or mislabelling may have contributed to

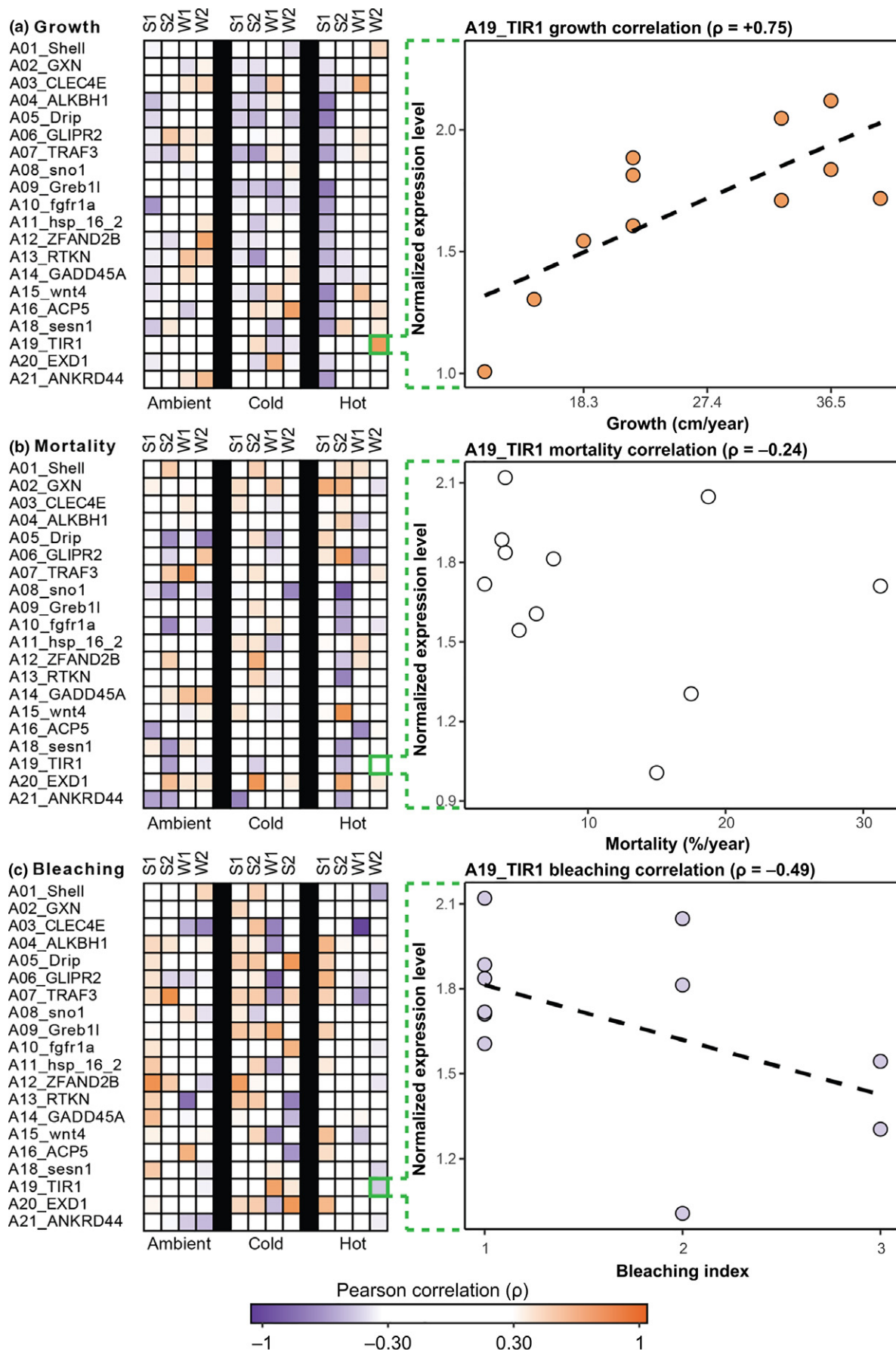


FIGURE 5 Relationships between host expression and colony performance. Heatmaps depict Pearson correlation coefficients (ρ) between coral holobiont phenotypes and expression levels in ambient, cold and hot temperature treatments at all seasons. Correlations are presented for host relative expression level and (a) annual growth rate, (b) annual mortality and (c) a bleaching frequency index ranging from 1 (mild) to 3 (severe). Plots of linear fits for the gene target A19_TIR1 in the hot treatment at Winter 2 (highlighted in green) illustrate a range of ρ values and the data that produce them. To aid visualization, relatively weak correlations ($|\rho| < .3$) appear white in the heatmaps [Colour figure can be viewed at wileyonlinelibrary.com]

these patterns, although we explicitly aimed to control these factors in the experimental design.

Alternatively, epigenetic modifications such as differential methylation of genes (Putnam, Davidson, & Gates, 2016) and/or somatic mutations (Devlin-Durante, Miller, Precht, & Baums, 2016; van Oppen, Souter, Howells, Heyward, & Berkelmans, 2011) among ramets of the same genet could explain large expression variation associated with a single coral genotype. In silico searches of the draft *A. cervicornis* genome (Kitchen, Baums, Fogarty, & Miller, 2018) recovered multiple homologs for genes that comprise the core methylation machinery. Environmental “memory” within individual colonies, likely driven in part by epigenetic modification, may persist for at least 10 years in corals (Brown, Dunne, Edwards, Sweet, & Phongsuwan, 2015). Therefore, branches of the same genet originally sourced from different parts of the donor colony may exhibit very different expression profiles reflecting past microhabitats (e.g., shaded vs. unshaded) prior to entering the nursery. This would mirror epigenetic and phenotypic patterns observed among ramets of clonal plants in different habitats (Douhovnikoff & Dodd, 2015). While the occurrence and accumulation of somatic mutations have now been documented in acroporid corals (Devlin-Durante et al., 2016; van Oppen et al., 2011), their potential effect on gene expression is unexplored. Whatever the cause, the sometimes large within-genet variation adds an extra layer of complexity when developing biomarkers for corals.

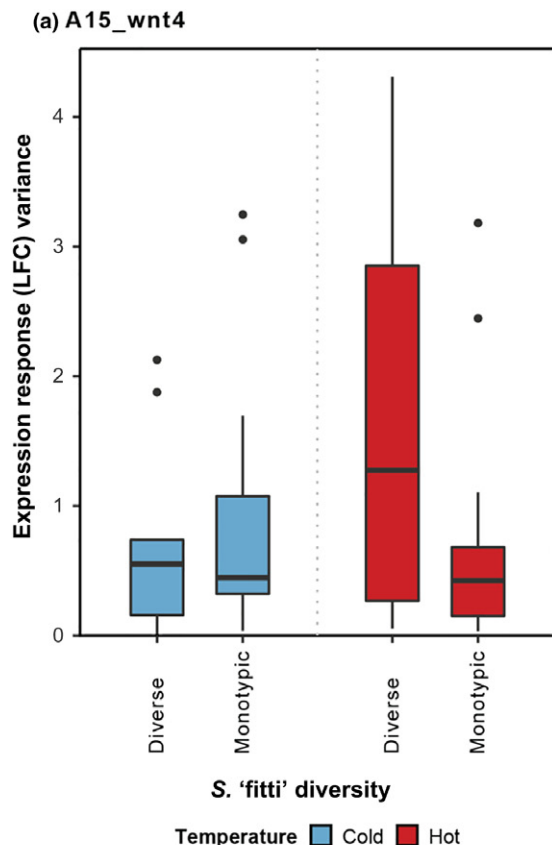
4.2 | Inconsistent correlations between gene expression and performance metrics

A primary goal of this study was to identify GEBs for holobiont performance. Despite finding some strong correlations between gene expression and growth rate, mortality and bleaching at different time points, few of the candidate biomarkers were consistently predictive of colony performance (Figure 5). The best candidate genes produced significant correlations with performance only in some of the seasons or with only some of the performance metrics, such that a strong predictor in Summer became a weak predictor in Winter, for example. Likewise, Bay and Palumbi (2017) recently identified two gene coexpression modules (large groups of genes that share similar expression patterns across treatments in an experiment) that significantly correlated with colony survival and/or growth in a reciprocal transplant study. One module showed consistent expression across multiple experiments (Rose, Seneca, & Palumbi, 2016), but the other did not, illustrating that high within-species variability in gene expression responses to stress are a general feature in corals.

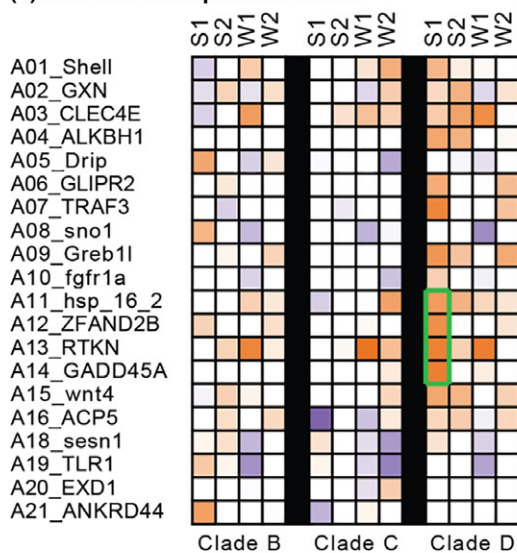
Development of gene expression markers for colony performance requires that good performance indicators are available. Performance measurements, let alone true lifetime fitness estimations, remain challenging in corals (Edmunds, 2017). Performance in the nursery was measured on the scale of a year (annual growth rates and mortality) or multiple years (bleaching). However, this study represented a narrow snapshot at four time points, and gene expression proved to be highly variable. These temporal incongruities likely made detection of correlations more challenging. Recently, calcification rate has been identified as a phenotype with strong genetic influence in *A. cervicornis* (Kuffner et al., 2017). Further development of high-throughput quantitative performance measures indicative of lifetime colony fitness should be a priority for coral research.

Our approach to identifying gene targets was conservative: first, we surveyed the entire transcriptome via pooled RNAseq, then we retained only functionally annotated genes for potential in-depth analysis. This meant that ~60% of host genes with interesting expression patterns were immediately disqualified from further consideration, even though unannotated genes can serve as useful biomarkers (e.g., Tang, Xiang, Terry, Kretzschmar, & Windl, 2010). For nonmodel organisms with relatively poor annotation like corals, there are arguments to be made for exploring unannotated genes in biomarker experiments. Knowing a gene's exact function is not necessary for the purpose of predicting performance, and the first step in determining an unknown gene's function could come from correlating its expression with interesting phenotypes.

Several actions could be taken to improve future efforts to identify predictive coral GEBs. The application of more sophisticated regression analyses, such as elastic nets, may uncover less obvious associations between gene expression and performance phenotypes. This and other coral studies would benefit from inclusion of a more direct measurement of coral host fitness and performance, but reliable phenotypes have proven elusive (Edmunds, 2017; Kuffner et al., 2017). Other short-term metrics, such as photosynthesis or respiration rates, could be measured simultaneously to assess the state of the symbionts. It will also be helpful to test less extreme, more gradually changing temperature stresses. Comparing stressors with different ramping speeds and magnitudes would be informative as well as useful for calibrating the amount of stress required to elicit an immediate gene expression change without causing mortality. This would allow researchers to extend the investigation to the recovery period after returning treated fragments to the nursery, taking full advantage of the experimental power of long-term in situ common garden studies. Finally, consistently up- or downregulated genes could be used in combination as the basis for “double-gene” assays (Kenkel et al., 2011).



(b) Normalized expression level



(c) Colony performance

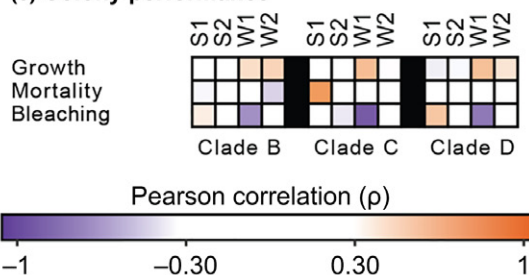


FIGURE 6 Influence of symbionts on host expression and colony performance. (a) Representative boxplots of variance in host expression responses (log2 fold changes; LFC) for target A15_wnt4 when comparing two groups of three colonies with different levels of intraspecific diversity in the dominant symbiont (*Symbiodinium* "fitti"). "Diverse" indicates three *S. "fitti"* strains (one per colony); "monotypic" indicates one *S. "fitti"* strain (shared in all three colonies). Symbiont strains were unique to each group. Also presented are heatmaps of Pearson correlation coefficients (ρ) between abundances of background symbionts (Clades B, C and D) and (b) host relative expression levels or (c) holobiont phenotypes at all seasons. Highlighted in green are four heat stress genes featuring high correlations with the abundance of Clade D (*S. trenchii*) during Summer 1. To aid visualization, relatively weak correlations ($|\rho| < .3$) appear white in the heatmaps [Colour figure can be viewed at wileyonlinelibrary.com]

4.3 | Candidate diagnostic biomarkers

Predicting performance is only one goal of developing GEBs; another is to detect stress. Although few genes correlated with colony performance metrics across different seasons (and therefore were not predictive), many responded consistently to temperature changes (and therefore were diagnostic). Four genes were particularly well suited to detecting heat stress in *A. cervicornis*. (A11_hsp-16.2, A12_ZFAND2B, A13_RTKN and A14_GADD45A; Figure 3c and Figure S1). All four are known members of stress pathways, coding for heat-shock, zinc finger, rhotekin and damage-inducible proteins, respectively. They were upregulated under hot conditions by nearly all genotypes regardless of nursery and varied only slightly by season. They were also members of the same gene coexpression module. The first principal component of all genes in the cluster (the eigengene) featured a high and statistically significant correlation with temperature ($\rho = +.57$; $p = .05$) but not season or nursery/location.

Four genes were consistently downregulated in response to both hot and cold stresses (A04_ALKBH1, A05_Drip, A18_sesn1 and A19_TIR1; Figure 3a and Figure S1). These genes encode alkylated DNA repair, aquaporin, sestrin and Toll-like receptor proteins, respectively. Given their similar expression patterns regardless of stress type, these genes may be useful for identifying general stresses such as pollution, disease and other acute impacts. These were all members of a separate gene coexpression module whose eigengene correlated with nursery/location ($\rho = -.66$; $p = .02$) but not temperature or season, suggesting they may be locally acclimatized/adapted (see below).

TNF receptor-associated factor 3 (A07_TRAF3) stood out as a potentially informative marker for seasonal heat acclimatization. This gene inhibits NF- κ B activation (Yamamoto, Kishimoto, & Minamoto, 1998), which is a key signalling component of the immune response and stress-induced apoptosis in humans (Gilmore & Wolenski, 2012) as well as corals (Davy, Allemand, & Weis, 2012; Zhou et al., 2017). In this study, the *A. cervicornis* TRAF3 homologue was typically downregulated during hot and cold exposures, likely activating NF- κ B and a subsequent stress response

(Figure 3d). However, TRAF3 was upregulated under heat shock during Summer 2 after the summer thermal maximum, reflecting a possible reduction in stress. This pattern suggests that the corals may have been “primed” by earlier heat exposure, making them more capable of dealing with heat stress in the late summer than any other period during the year.

4.4 | Local adaptation

By design, our HTqPCR analysis focused on a subset of genes that appeared to have similar expression patterns across nurseries (at least according to the initial RNAseq results). The goal was to identify universal biomarkers for *A. cervicornis*, and so we tried to filter out genes with nursery-specific patterns. Nevertheless, there was a large nursery signal in the RNAseq data set, as well as among some of the HTqPCR genes (Figures 1d and 4). This suggests differential acclimatization among corals in different nurseries. It might also indicate a degree of local adaptation across the latitudinal gradient within Florida.

Local adaptation despite gene flow has been inferred in other Caribbean species (Kenkel, Meyer, & Matz, 2013; Polato et al., 2010), and *A. cervicornis* genotype, environment and their interaction greatly impact colony growth, survivorship and tolerance in the Florida Keys (Drury, Manzello, & Lirman, 2017). Reciprocal transplant studies have demonstrated differential acclimatization as well as local adaptation with respect to coral gene expression, indicating transcriptional plasticity itself may be an adaptive trait serving as a genomic basis for resilience to climate change (Barshis et al., 2013; Kenkel & Matz, 2016; Palumbi, Barshis, Traylor-Knowles, & Bay, 2014). *A. cervicornis* is an ideal system to further explore these concepts through transcriptional comparisons between the marginal Florida population and more central Caribbean populations. Additional work will be required to test the feasibility of using locally adapted genes as expression biomarkers of colony performance.

4.5 | Opposing versus allied gene expression responses

One previous study on tropical reef-building corals quantified levels of a thermal stress response protein in the same individuals during hot and cold stress, finding a strong increase in protein abundance under both treatments after 6 hr (Seveso et al., 2016). Such allied molecular responses are expected to reflect the ability of an organism to respond to multiple stressors using a common core stress response. In contrast, opposing responses suggest the organism requires unique mechanisms to handle different stressors. In the context of threatened corals, a greater ratio of allied to opposing thermal stress response genes in a given colony could indicate greater resilience to future sea surface temperature fluctuations caused by climate change, with implications for the likelihood of poleward range extension. Only three genes of interest truly featured opposing patterns. Although some allied genes also turned out to be false positives, the low proportion of opposing genes ($n = 10$)

relative to the high number of allied genes ($n = 58$) from the RNA-seq experiment indicates that it is more common for coral stress genes to act in the same direction than to be opposed under different thermal shocks (Figure 1e), and these allied responses tend to be consistent across seasons (e.g., Figure 3a).

Even though hot and cold stresses generally affect different molecular pathways, certain elements of the response appear to be conserved in *A. cervicornis*. It is also more common for coral stress genes to be downregulated together ($n = 56$) than to be upregulated together ($n = 2$) during alternate stress events (Figure 1e). Therefore, this *A. cervicornis* population does not appear to be constrained to intermediate expression of stress response genes despite exposure to different thermal extremes during Summer and Winter. Rather, alternate stressors may reinforce the expression patterns of genes shared in both responses. It is possible that any Florida host genotypes showing opposing gene expression patterns with hot and cold stress have died out during the past few decades or that such genotypes do not exist in this species. It thus would be interesting to investigate *A. cervicornis* gene expression patterns in more central locations in the Caribbean with narrower temperature ranges.

4.6 | No clear influence of dominant or background symbiont diversity on host gene expression and performance

Given that selection can act at the level of the coral holobiont (Iglesias-Prieto & Trench, 1997; Lajeunesse et al., 2010; Parkinson & Baums, 2014), we posited that maintaining genetic diversity of both hosts and symbionts—not just the host—should be an important goal in the conservation of endangered corals. We therefore tested whether dominant or background symbiont composition influenced host expression or holobiont phenotypes. However, we found no strong evidence for such an effect.

The dominant *S. “fitti”* strain was no more likely to predict host expression levels than host genotype (Figure 4), and host expression response variance was just as high among groups of colonies sharing a single *S. “fitti”* strain as it was among groups of colonies with multiple strains (Figure 6a). Moreover, correlations between background symbiont composition and host expression were mostly weak and variable (Figure 6b). Of note, background Clade D abundance, which in the Caribbean is represented by *S. trenchii* (= ITS2 type D1a), strongly correlated with the ambient expression levels of four candidate heat-shock proteins during Summer 1, when temperatures in the Florida Keys were rapidly climbing. Additionally, all three background clade abundances correlated negatively with longer term bleaching frequency during Winter 2 (Figure 6c). At this point, it is unclear whether these were spurious associations or meaningful biological interactions, but they warrant further investigation.

It is possible that symbiont composition did not affect colony performance at all, but it is more likely that host expression is simply a poor metric by which to measure a symbiont genotype and/or species effect during acute thermal stress. Host gene expression

changes are rapid and dynamic, whereas *Symbiodinium* appear far less transcriptionally responsive to temperature shocks (Leggat, Seneca, et al., 2011; Barshis, Ladner, Oliver, & Palumbi, 2014; this study; but see McGinley et al., 2012; Baumgarten et al., 2013; Levin et al., 2016). This is likely due to the large extent of posttranscriptional regulation in dinoflagellates, along with their other unique genomic features such as permanently condensed chromosomes (Leggat, Yellowlees, & Medina, 2011; Morse, Milos, Roux, & Hastings, 1989; Murray et al., 2016). Additionally, a “host buffering” effect may temporarily protect symbiont cells from rapid changes in the host environment (Parkinson et al., 2015). Given the different time frames over which coral hosts and their algal symbionts experience and respond to stress at a molecular level, transient shocks appear insufficient to detect genotypic interactions among host–symbiont partners. Longer term exposures to less extreme temperatures or alternate metrics such as proteomics, metabolomics or cellular physiological assays should be used to further characterize the phenotypes of these fine-scale interactions and assess the conservation priority of symbiont diversity.

4.7 | Broader implications

All GEBs in this study were designed specifically for *A. cervicornis*, but they are likely to work in other acroporids as well. For example, Kenkel et al. (2011) showed that at least two markers designed for the coral *Porites astreoides* could also be used effectively in congeneric *P. lobata* without redesigning the primers. In a brief test, we BLASTed our target *A. cervicornis* transcripts against genomic and transcriptomic data sets from 20 reef-building coral species hosted at reefgenomics.org (Bhattacharya et al., 2016). Among the five acroporid species, sequences typically shared >95% identity for stretches exceeding 500 base pairs in length (data not shown). More distantly related species also had relatively high similarity scores (often > 70%). Thus, the assays could be useful for a broad range of corals, although further experimentation would be required.

Incorporating GEBs into restoration programmes could take many forms. GEBs could be used as a litmus test to compare the relative tolerances of wild colonies: this would allow nurseries to balance resource investment between propagating resilient genotypes and safeguarding more vulnerable genotypes (perhaps by moving them to cooler locations). An added benefit of the developed array is that it can be used to monitor changes in background symbiont populations at the same time as taking the “pulse” of the host.

Beyond corals, our methodology could be used to identify and test biomarkers in any nonmodel system. The combined RNAseq and HTqPCR approach, while expensive, provided many diagnostic gene targets along with a wealth of expression data useful for both health assessment and biological investigation. Similar strategies may prove useful in other species of conservation value, particularly when considering adaptive potential in the light of global change. It will be important to examine the effects of opposing stressors in organisms likely to be impacted by a more variable climate. These effects will

need to be tested multiple times throughout the year, as environmental extremes fluctuate seasonally. Finally, the transcriptional variability in our coral system cautions that even highly clonal organisms require extensive replication to arrive at representative gene expression values for a given phenotype.

5 | CONCLUSION

We have improved on previous efforts to identify GEBs for corals by developing new high-throughput methods, interrogating a large number of host genes simultaneously, expanding the number of individuals assessed, repeating the experiments at multiple time points throughout a year and incorporating *Symbiodinium* diversity. Working with nursery-reared corals reduced the influence of environmental variation and ensured that our results produced useful information tied directly to ongoing restoration projects. Despite finding few consistent correlations between ambient and temperature shock expression variation, holobiont performance and symbiont diversity, we nevertheless identified several potentially useful diagnostic thermal stress biomarkers. Signs of allied expression patterns under hot and cold events across markers imply that *A. cervicornis* is not restricted to an intermediate response despite seasonally fluctuating selective pressures. Although challenging for GEB development, the high levels of standing transcriptional variation observed among individuals in this study suggest restoration nurseries are fulfilling their role as repositories for coral genetic and phenotypic diversity.

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DATA ACCESSIBILITY

The RNAseq data are deposited in the NCBI GEO database (Accession no. GSE108338). All raw data, R code and additional supplements associated with this manuscript can be accessed in the Pennsylvania State University's SCHOLARSPIRE database (<https://doi.org/10.18113/s1rp4r>). Additional supplements include RNAseq library composition, Illumina run statistics, RNAseq differential expression results (statistics, biological coefficient of variation, multidimensional scaling, mean

expression vs. variance, and volcano plots), updated *A. cervicornis* gene transcript annotations, HTqPCR assay design, complete correlation outputs, graphs of individual genotype variation for each gene and graphs of “diverse” versus “monotypic” variance for each gene.

AUTHOR CONTRIBUTIONS

J.E.P., D.L., T.C.L. and I.B.B. conceived of this project. J.E.P., E.B., C.L., K.N. and S.S. performed field experiments. J.E.P. and M.K.D. performed laboratory experiments. J.E.P. analysed the data and created the figures. J.E.P. wrote the manuscript. All authors contributed editorially to the final manuscript.

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SUPPORTING INFORMATION

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